Prasanta Char

Curriculum Vitae

Departamento de Física Fundamental,
Facultad de Ciencias, Universidad de Salamanca,
Plaza de la Merced S/N, 37008 Salamanca, Spain

Personal Information

Date of Birth :	**27 October, 1987**.	Address :	**Calle Bretón 16, 2D**.
Nationality :	**Indian**.		**37001 Salamanca, Spain**.
Sex :	**Male**.	E-mail :	**prasanta.char@usal.es, prasantachar@gmail.com**.
INSPIRE ID:	**P.Char.1**.		
ORCID ID:	**0000-0001-6592-6590**.	Cell Phone :	**+34-638923175**.

Current Position

Marie Skłodowska-Curie Postdoctoral Fellow, Universidad de Salamanca, Spain

Research Interests

Neutron stars; dense nuclear matter; relativistic mean field theory; hyperons and quark matter; tidal deformability; multimessenger inference; asteroseismology; gravitational wave parameter estimation; Bayesian methods; superfluidity; pulsar glitches; neutron star cooling

Education

2012-2016 **Ph.D**, *Astroparticle Physics & Cosmology Division, Saha Institute of Nuclear Physics*, Kolkata, India.

Thesis Title: **Some Studies on Core-Collapse Supernovae and Neutron Stars**, *degree awarded by*, Homi Bhabha National Institute, Mumbai, India, degree awarded on **25/04/2017**.

2011–2012 **Post M.Sc. Associateship Course (Physics)**, *Saha Institute of Nuclear Physics*, Kolkata, India.

2009–2011 **M.Sc. (Physics)**, *University of Calcutta*, Kolkata, India.

2006–2009 **B.Sc. (Physics Honours)**, *Narasinha Dutt College, University of Calcutta*, Kolkata, India.

Awards

2023 **Awarded**, *USAL4EXCELLENCE (MSCA-COFUND) Fellowship 2022* , started in 2024.

2023 **Awarded**, *Marie Skłodowska-Curie Postdoctoral Fellowship 2022 (HORIZON-MSCA-2022-PF-01).*

2023 **Awarded**, *WINNINGNormandy (MSCA-COFUND) Fellowship 2022* , declined.

2020 **Awarded**, *IISN-FNRS Postdoctoral Fellowship*, at the University of Liege.

2018 **Awarded**, *INFN Postdoctoral Fellowship in Theoretical Physics*, by Istituto Nazionale di Fisica Nucleare (INFN).

2010 **Qualified**, *National Eligibility Test (Joint CSIR-UGC NET December 2010)*, conducted by Council of Scientific and Industrial Research, India and awarded a Research Fellowship.

Research Experience

09/2023-present **Postdoctoral Fellow**, at Universidad de Salamanca, Spain.

09/2020-08/2023 **Postdoctoral Fellow**, at Université de Liège, Belgium.

09/2018-08/2020 **Postdoctoral Fellow**, at Istituto Nazionale di Fisica Nucleare (INFN) Sezione di Ferrara, Italy.

01/2017-08/2018 **Postdoctoral Fellow**, at Gravitational Wave Data Centre, Inter-University Centre for Astronomy and Astrophysics , Pune, India.

08/2012-08/2016 **PhD Research Fellow**, at Astroparticle Physics & Cosmology Division, Saha Institute of Nuclear Physics, Kolkata, India.

Publications

Journal Articles

1. *Systematic study of scalar, vector, and mixed density dependencies in relativistic mean-field descriptions of hyperonic matter in neutron stars*
   Aprajita Shrivastava, Prasanta Char, Sakshi Gautam, Sarmistha Banik
   arXiv:2511.01505, **under review.**
2. *Implication of multimessenger observations on the relativistic mean-field equation of state of dense nuclear matter and skin thickness of nuclei*
   Rahul Kumar, Prasanta Char, Rana Nandi
   arXiv:2511.00646, **under review.**
3. *Properties of Neutron Stars with Hyperons within a Relativistic Metamodel*
   Prasanta Char, Chiranjib Mondal, Timothé Alezraa, Francesca Gulminelli, Micaela Oertel
   arXiv:2510.00997, **under review.**
4. *Systematics from NICER Pulse Profiles Drive Uncertainty in Multi-Messenger Inference of the Neutron Star Equation of State*
   Bhaskar Biswas, Prasanta Char
   arXiv:2507.12540, **Phys. Rev. D 112 (2025) 6, 063049**
5. *Stellar properties indicating the presence of hyperons in neutron stars*
   Andreas Bauswein, Aristeidis Nikolaidis, Georgios Lioutas, Hristijan Kochankovski, Prasanta Char, Chiranjib Mondal, Micaela Oertel, Laura Tolos, Nicolas Chamel, Stephane Goriely
   arXiv:2507.10372, **under review.**
6. *Exploring the limits of nucleonic metamodeling using different relativistic density functionals*
   Prasanta Char, Chiranjib Mondal
   arXiv:2502.04211, **Phys. Rev. D 111 (2025) 10, 103024**
7. *Gauge-invariant perturbations of relativistic non-perfect fluids in spherical spacetime*
   David Díaz-Guerra, Conrado Albertus, Prasanta Char, M. Ángeles Pérez-García
   arXiv: 2410.18081, **Phys. Rev. D 110 (2024) 12, 124012**
8. *Structural response of neutron stars to rapid rotation and its impact on the braking index*

Avishek Basu, Prasanta Char, Rana Nandi
arXiv:2409.11558, **Phys. Rev. D 112 (2025) 2, 023020**

9. *The compact object of HESS J1731-347 and its implication on neutron star matter*
Prasanta Char, Bhaskar Biswas
arXiv:2408.15220, **under review.**
10. *Radial Oscillations of Neutron Stars with Antikaon Condensate*
Apurba Kheto, Prasanta Char
arXiv:2308.07995, **Eur. Phys. J. A 59 (2023) 9, 201**
11. *Generalised description of Neutron Star matter with nucleonic Relativistic Density Functional*
Prasanta Char, Chiranjib Mondal, Francesca Gulminelli, Micaela Oertel
arXiv:2307.12364, **Phys. Rev. D 108 (2023) 10, 103045**
12. *Stability and instability of strange dwarfs*
Francesco Di Clemente, Alessandro Drago, Prasanta Char, Giuseppe Pagliara
arXiv:2207.08704, **Astron. Astrophys. 678 (2023) L1**
13. *Speed of sound in dense matter and two families of compact stars*
Silvia Traversi, Prasanta Char, Giuseppe Pagliara, Alessandro Drago
arXiv:2102.02357, **Astron. Astrophys. 660 (2022) A62**
14. *GW190814: on the properties of the secondary component of the binary*
Bhaskar Biswas, Rana Nandi, Prasanta Char, Sukanta Bose, Nikolaos Stergioulas
arXiv:2010.02090, **Mon. Not. Roy. Astron. Soc. 505 (2021) 2, 1600-1606.**
15. *Towards mitigation of apparent tension between nuclear physics and astrophysical observations by improved modeling of neutron star matter*
Bhaskar Biswas, Prasanta Char, Rana Nandi, and Sukanta Bose,
arXiv:2008.01582 , **Phys. Rev. D 103 (2021) 10, 103015.**
16. *Structure of Quark Star: A Comparative Analysis of Bayesian Inference and Neural Network Based Modeling*
Silvia Traversi, Prasanta Char
arXiv:2007.10239, **Astrophys. J. 905 (2020) 1, 9**
17. *Bayesian Inference of Dense Matter Equation of State within Relativistic Mean Field Models using Astrophysical Measurements*
Silvia Traversi, Prasanta Char, Giuseppe Pagliara,
arXiv:2002.08951, **Astrophys. J. 897 (2020) 2, 165**
18. *Observed glitches in eight young pulsars*
Avishek Basu, Bhal Chandra Joshi, M.A. Krishnakumar, Dipankar Bhattacharya, Rana Nandi, Debades Bandyopadhyay, Prasanta Char, P.K. Manoharan,
arXiv:1911.04934, **Mon. Not. Roy. Astron. Soc. 491 (2020) 3, 3182-3191**.
19. *Effect of superfluid matter of neutron star on tidal deformability*
Sayak Datta, Prasanta Char,
arXiv:1908.04235, **Phys. Rev. D 101 (2020) 6, 064016**.
20. *Role of crustal physics in the tidal deformation of a neutron star*
Bhaskar Biswas, Rana Nandi, Prasanta Char, Sukanta Bose,
arXiv:1905.00678, **Phys. Rev. D 100 (2019) 4, 044056**.
21. *Constraining the relativistic mean-field model equations of state with gravitational wave observations*
Rana Nandi, Prasanta Char, Subrata Pal,
arXiv:1809.07108, **Phys. Rev. C 99 (2019) 5, 052802**.

22. *Relativistic tidal properties of superfluid neutron stars*
Prasanta Char, Sayak Datta,
arXiv:1806.10986, **Phys. Rev. D 98 (2018) 8, 084010**.
23. *Constraining strangeness in dense matter with GW170817*
R.O. Gomes, Prasanta Char, S. Schramm,
arXiv:1806.04763, **Astrophys. J. 877 (2019) 2, 139**.
24. *Glitch Behavior of Pulsars and Contribution from Neutron Star Crust*
Avishek Basu, Prasanta Char, Rana Nandi, Bhal Chandra Joshi, Debades Bandyopadhyay,
arXiv:1806.01521, **Astrophys. J. 866 (2018) 2, 94** .
25. *Properties of Massive Rotating Protoneutron Stars with Hyperons: Structure and Universality*
Smruti Smita Lenka, Prasanta Char, Sarmistha Banik,
arXiv:1805.09492, **J. Phys. G 46 (2019) 10, 105201**.
26. *Hybrid stars in the light of GW170817*
Rana Nandi, Prasanta Char,
arXiv:1712.08094, **Astrophys. J. 857 (2018) 1, 12**.
27. *Moment of inertia, quadrupole moment, Love number of neutron star and their relations with strange matter equations of state*
Debades Bandyopadhyay, Sajad A. Bhat, Prasanta Char, Debarati Chatterjee,
arXiv:1712.01715, **Eur. Phys. J. A 54 (2018) 2, 26**.
28. *Critical Mass, Moment of Inertia and Universal Relations of Rapidly Rotating Neutron Stars with Exotic Matter*
Smruti Smita Lenka, Prasanta Char, Sarmistha Banik,
arXiv:1704.07113, **Int. J. Mod. Phys. D 26 (2017) 11, 1750127**.
29. *Role of Nuclear Physics in Oscillations of Magnetars*
Rana Nandi, Prasanta Char, Debarati Chatterjee, and Debades Bandyopadhyay,
arXiv:1608.01241, **Phys. Rev. C 94 (2016) 2, 025801**.
30. *A Comparative Study of Hyperon Equations of State in Supernova Simulations*
Prasanta Char, Sarmistha Banik, Debades Bandyopadhyay,
arXiv:1508.01854, **Astrophys. J. 809 (2015) 2, 116**.
31. *Massive Neutron Stars with Antikaon Condensates in a Density Dependent Hadron Field Theory*
Prasanta Char, Sarmistha Banik,
arXiv:1406.4961, **Phys. Rev. C 90 (2014) 1, 015801**.

Journal Articles: Collaborations

1. *Probing neutron star interiors and the properties of cold ultra-dense matter with the SKA*
Avishek Basu et al. (The SKA Pulsar Science Working Group),
arXiv:2512.16162, **Open Journal of Astrophysics, Vol. 8 Suppl., 1, 2025 (doi:10.33232/001c.154253)**.
2. *The Science of the Einstein Telescope*
Adrian Abac et al. (Einstein Telescope Collaboration),
arXiv:2503.12263, **Accepted in the Journal of Cosmology and Astroparticle Physics**.
3. *Tests of General Relativity with GW230529: a neutron star merging with a lower mass-gap compact object*
Elise M. Sänger et al.,
arXiv:2406.03568, **Under review**.
4. *Observation of Gravitational Waves from the Coalescence of a $2.5-4.5M_\odot$ Compact Object and a Neutron Star*
LIGO Scientific and Virgo, and KAGRA and VIRGO Collaborations,

arXiv: 2404.04248, **Astrophys. J. Lett. 970 (2024) 2, L34**.

Professional Experience

Reviewing in Internationally Reputed Journals

Astrophysical Journal, Physical Review D, Physical Review C, Physical Review Letters, European Physical Journal A, European Physical Journal C, Monthly Notices of Royal Astronomical Society, Physics of the Dark Universe, Journal of High Energy Astrophysics

Major Collaborations

2020-present **Virgo Collaboration**, member of gravitational wave observational scientific collaboration.

2022-present **Einstein Telescope Collaboration**, member of OSB Division 6 for Nuclear Physics. Contributed to the ET Bluebook chapter on subatomic physics.

2024-present **Square Kilometre Array Observatory**, member of pulsar sciences working group. Contributed to the EOS with SKA paper, soon to be published.

2025-present **NewAthena Science Study Team for Athena X-ray observatory**, member of Working Group 4 for Compact Objects.

Supervising Students

2018-2019 **Andrea Pavan**, *Masters student, University of Padova*, supervised jointly with Prof. Alessandro Drago and Prof. Giuseppe Pagliara.

Teaching and Mentoring

In 2024, I have taught a 20 hour course of nuclear astrophysics to bachelor students with 2 ECTS credit points at the University of Salamanca. I have frequently collaborated with and mentored PhD students informally in my past groups in India and Italy, providing them regular supervision during their theses, and helping with their career paths.

Outreach and organizational activities

I participated in outreach activities as a part of the BIE network at USAL, Bachillerato Excelencia for secondary school at IES La Vaguada. I helped to organize the Iberian Gravitational Wave meeting 2024 as a member of the local organizing committee.

Talks given in Conferences

1. **Exploring Composition of Neutron Star Matter with Astrophysical Observations** 21st conference in the series "Recent Developments in Gravity" (NEB-21), September 1-4, 2025, Corfu, Greece.
2. **Exploring Composition of Neutron Star Matter with a Relativistic Density Functional** 2nd Workshop on modern equations of state and spectroscopy in neutron-star matter, May 26-28, 2025, Alcalá de Henares, Madrid, Spain
3. **Exploring Composition of Neutron Star Matter with a Relativistic Density Functional** CoCoNuT Meeting, December 11-13, 2024, Valencia, Spain
4. **Exploring Composition of Neutron Star Matter with Astrophysical Observations** 1st TEONGRAV international workshop on theory of gravitational waves, September 16-20, 2024, Sapienza University of Rome, Italy
5. **Exploring Composition of Neutron Star Matter with Astrophysical Observations** XIIth International Symposium on Nuclear Symmetry Energy, September 9-13, 2024, Caen, France

6. **Exploring Composition of Neutron Star Matter with a Relativistic Density Functional** 10th International Conference on Quarks and Nuclear Physics (QNP2024), July 8-12, 2024, Barcelona, Spain
7. **Exploring Composition of Neutron Star Matter with Astrophysical Observations** 13th Iberian Gravitational Waves Meeting June 24-26, 2024, Salamanca, Spain
8. **Relativistic description of neutron star matter with latest experimental and observational constraints** 19th International Conference on QCD in Extreme Conditions (XQCD 2023) July 26-28, 2023, Department of Physics (University of Coimbra), Portugal
9. **Modelling the dense nuclear matter equation of state consistent with the astrophysical observations** Advances in Astroparticle Physics and Cosmology (AAPCOS-2023), January 23-27, 2023, Saha Institute of Nuclear Physics, Kolkata, India
10. **Modelling Neutron Star Matter** Belgian-Dutch Gravitational Wave Meeting 2022, October 13-14, 2022, Ghent, Belgium
11. **Constraints on Dense Matter Equation of State from Astrophysical Observation** X International Symposium on Nuclear Symmetry Energy, September 26-30, 2022, Catania, Italy
12. **Dense Matter Equation of State from Multimessenger Observations** One day Symposium on Neutron Stars, June 16, 2021, BITS-Pilani, Hyderabad, India
13. **Properties of the secondary component of GW190814** Belgian Gravitational Wave Meeting, October 27, 2020, Brussels, Belgium
14. **Constraints on Dense Matter Equation of State from GW170817: Possibility of Different Families of Compact Stars** Neutron stars and their environments (MODE-SNR-PWN), April 8-10, 2019, Orléans, France
15. **Effect of superfluidity in neutron star in two-fluid scenario** Advances in Astroparticle Physics and Cosmology (AAPCOS-2018), March 6-9, 2018, Saha Institute of Nuclear Physics, Kolkata, India
16. **Tidal Deformability of Compact Stars with Quark Hadron Phase Transition** 29^{th} meeting of the Indian Association for General Relativity and Gravitation (IAGRG), May 18-20, 2017, Indian Institute of Technology, Guwahati, Assam, India
17. **Comparison of Hyperonic Equations of State for Core Collapse Supernovae Simulations** Recent Trends in the study of Compact Objects – Theory and Observations (RETCO-II), May 6-8, 2015, Aryabhatta Research Institute of Observational Science, Nainital, India
18. **New Hyperon Equation of State table for Supernova Simulations and Neutron Stars** 7^{th} International Conference on Physics & Astrophysics of Quark Gluon Plasma, February 2-6, 2015, Variable Energy Cyclotron Centre, Kolkata, India

Invited Seminars

1. **Exploring the Composition of Neutron Star Matter with Astrophysical Observations** at the Department of Astrophysics, Geophysics and Oceanography, Université de Liège, Liège, Belgium, on November 5, 2024
2. **Generalized description of neutron star matter with a relativistic density functional** at IAA Seminar online, ULB, Brussels, on June 11, 2024
3. **Generalized description of neutron star matter with a relativistic density functional** at NP3M Seminar online, on April 11, 2024
4. **Probing Neutron Star Matter Using Astrophysical Observations** at Gran Sasso Science Institute, L'Aquila, Italy, on November 3, 2021
5. **Probing Neutron Star Matter Using Astrophysical Observations** at GANIL, Caen, France, on September 17, 2021

6. **Dense Matter inside Neutron Stars: Constraints from Astrophysical Observations** at the Department of Astrophysics, Geophysics and Oceanography, Université de Liège, Liège, Belgium on December 3, 2020
7. **Tidal deformability of neutron stars with superfluid interior** at the Center of Astrophysics and Gravitation (CENTRA), Instituto Superior Técnico (IST) on October 10, 2019
8. **Effect of superfluidity in neutron stars in a two-fluid scenario** at the Tata Institute of Fundamental Research (TIFR), Mumbai, India, on April 12, 2018
9. **Core Collapse Supernova: Role of Hyperonic Matter** at the International Centre for Theoretical Sciences (ICTS-TIFR), Bengaluru, India on March 1, 2017
10. **Role of hyperon matter in core collapse supernova** at the European Centre for Theoretical Studies in Nuclear Physics and Related Areas (ECT*), Trento, Italy on June 9, 2016
11. **Effects of Hyperonic matter in Core Collapse Supernova** at the Inter-University Centre for Astronomy and Astrophysics, on April 26, 2016

References

Dr. **Micaela Oertel**,
Senior Research Scientist
Observatoire Astronomique de Strasbourg,
11 rue de l'Université - 67000 Strasbourg, France
E-mail : *micaela.oertel@astro.unistra.fr*.

Prof. **Giuseppe Pagliara**,
Professor
Department of Physics, University of Ferrara and INFN Sezione di Ferrara, Via Saragat 1, 44122 Ferrara, Italy
E-mail : *pagliara@fe.infn.it*.

Dr. **Rana Nandi**,
Assistant Professor
Department of Physics, Indian Institute of Technology Delhi,
New Delhi 110016, India
E-mail : *rananandi@iitd.ac.in*.

Prof. **Sarmistha Banik**,
Professor
Department of Physics, BITS Pilani, Hyderabad Campus
Shameerpet Mandal, Hyderabad 500078
E-mail : *sarmistha.banik@hyderabad.bits-pilani.ac.in*.